

Madhu Biradakota

United States | Phone: +19284237367 | Email: madhubiradakota01@gmail.com | [LinkedIn](#)

SUMMARY

- Data Engineer with 6+ years of expertise in designing cloud-native, large-scale data ecosystems that power analytics, automation, and machine learning across Robotics, E-commerce, and Scientific research domains.
- Skilled in architecting real-time and batch data pipelines using AWS (Glue, Redshift, Lambda, MWAA), Spark, Airflow, and Snowflake, processing 10TB+ daily with sub-minute latency and 99.9% reliability.
- Delivered measurable business impact by reducing infrastructure costs by 30%, cutting ETL latency by 45%, and translating complex research problems (Ranavirus outbreak prediction, robotic delivery optimization) into production-grade solutions.
- Engineer intelligent data systems that enable self-service analytics, ML-ready architectures, and data democratization, empowering teams to experiment and make data-driven decisions at scale.
- Excel in high-velocity startup environments where rapid iteration, cross-functional collaboration, and technical innovation drive business outcomes.

EXPERIENCE

Data Engineer

Amazon Robotics, Seattle

Oct 2024 – Present

- Architected and optimized large-scale, cloud-native data warehousing and analytics platforms supporting Amazon's Robotics and Operations Science teams, enabling predictive automation, operational intelligence, and ML experimentation.
- Decommissioned legacy Oracle systems by rebuilding 250+ SQL workflows in SparkSQL and streamlining schema inference, successfully migrating 10 TB+ daily operational data to Amazon Redshift, reducing compute cost by 30% and improving query latency by 45%.
- Orchestrated a fully serverless pipeline leveraging Airflow DAGs, Amazon S3, and AWS Lambda functions that dynamically infer Iceberg table schemas from DynamoDB metadata, ensuring zero manual intervention.
- Developed MWAA-orchestrated Glue pipelines incorporating ETL/ELT design and data modeling best practices, achieving 99.9% pipeline reliability with full data lineage tracking and standardized IaC deployments via CloudFormation and CDK.
- Built telemetry ingestion frameworks for robotics warehouse data, enabling near-real-time insights into throughput, path efficiency, and equipment anomalies, cutting decision latency by 60%.
- Collaborated with 8+ Data Scientists and ML Engineers across Robotics Operations to operationalize predictive models optimizing warehouse layout and robotic scheduling, improving fulfillment throughput by 12%.
- Introduced data quality automation via Python, Pandas, and AWS Glue Crawlers, detecting and reconciling anomalies across 50+ data sources and increasing trust in robotics analytics data by ~35%.

Research Data Engineer

Northern Arizona University, Arizona

Mar 2023 – Oct 2024

- Collaborated with cross-disciplinary biologists and data scientists to model and predict Ranavirus spread in amphibian populations using cloud-based data engineering and ML techniques.
- Constructed production-grade ETL pipelines on AWS EMR and S3 processing 2TB+ of ecological, climatic, and genomic datasets, managed via AWS Step Functions with automated retry logic and monitoring, enabling reproducible experimentation for 4 research teams.
- Created scalable feature pipelines and model training infrastructure using Python (Scikit-learn, Pandas, NumPy), enabling researchers to forecast Ranavirus outbreaks across 200+ habitat sites with 92% accuracy.
- Unified geospatial and temporal datasets (climate, pond proximity, migration patterns) to enhance predictive precision, decreasing false-positive outbreak predictions by 30%.
- Automated data cleaning, transformation, and validation workflows using Python and AWS Step Functions, processing data from 15+ research sources with 40% improvement in pipeline reliability and data readiness.
- Deployed visualization dashboards using Tableau for ecological scientists to monitor virus spread simulations and intervention outcomes.

Cloud Data Engineer

Saks, India

Jul 2021 – Dec 2022

- Architected and delivered a cloud-native data platform for an e-commerce ecosystem supporting real-time sales analytics, customer insights, and operational intelligence at scale.
- Designed a centralized data lake using AWS S3, Glue, and Redshift, integrating 40+ transactional, clickstream, and vendor data sources to provide a single source of truth across business units.
- Implemented Spark and Airflow-based ETL pipelines to coordinate ingestion and transformation of multi-terabyte data workloads, improving processing efficiency by 55% and cutting refresh times from 8 hours to under 30 minutes.
- Launched Kafka-based real-time streaming pipelines processing 50M+ events daily to capture user behavior and order events, powering recommendation systems and fraud detection engines with sub-10-second latency.
- Enhanced Snowflake queries through intelligent partitioning, compression, and clustering strategies, lowering compute cost by 35% and improving dashboard response times.
- Created a self-service analytics layer using Databricks and AWS QuickSight, empowering 50+ data analysts and business users across 6 departments to create custom KPIs and reports, reducing engineering ticket volume by 65%.
- Integrated CI/CD pipelines via Jenkins for continuous deployment of Glue jobs and schema migrations, ensuring consistent environment provisioning.

Data Engineer

Zappos, India

Nov 2018 – Jun 2021

- Contributed to a unified metrics platform supporting 40+ analysts across 8 countries, powering real-time business intelligence for \$2B+ annual e-commerce operations.
- Engineered end-to-end ETL pipelines on Hadoop, Spark, and Hive, processing 5 TB+ daily from transactions, marketing, and inventory systems to power business dashboards.
- Built a self-service metrics framework enabling 50+ business users to define custom KPIs via SQL templates, reducing deployment time from 2 weeks to 6 hours.
- Developed modular data ingestion workflows using Oozie and Unix Shell scripts, improving reliability and minimizing manual maintenance by 60%.
- Boosted Spark jobs and Hive queries through partitioning and indexing, improving pipeline performance by 45% and cutting cluster costs by 25%.
- Established reusable data transformation libraries and standardized schema management for retail datasets, ensuring consistent data governance and quality.
- Partnered with 10+ data scientists and analysts to deliver model-ready datasets for recommendation engines serving 5M+ daily users, personalization, and pricing optimization.
- Streamlined deployments through Git-based CI/CD pipelines and job automation, cutting release overhead by 30% while improving version control and rollback safety.

SKILLS

- **Cloud Services:** AWS (EC2, EMR, Glue, S3, Athena, Redshift, DynamoDB, Kinesis, Lambda, Step Functions, CloudFormation)
- **Frameworks and Processing:** Hadoop, Spark, Flink, NiFi, Airflow, DBT, MapReduce, Hive
- **Data Management:** ETL/ELT, Data Modeling, Data Warehousing, Data Lakehouse Architecture, Data Governance
- **Programming Languages:** Python, Java, Scala, SQL, Unix Shell, R, C/C++, C#
- **DevOps and IaC:** Jenkins, Terraform, Ansible, CloudFormation, CDK, Git, Maven, Bitbucket
- **Streaming & Integration:** Kafka, CDC, Oozie, Talend, Informatica
- **Databases and Storage:** Snowflake, Redshift, PostgreSQL, MySQL, MongoDB, Cassandra, Oracle, Cosmos DB
- **Data Analysis:** Tableau, Power BI, Databricks, AWS QuickSight
- **Certifications:** AWS Certified Data Engineer – Associate, AWS Certified Cloud Practitioner

EDUCATION

Northern Arizona University, Arizona

Jan 2023 -May 2024

Master of Science in Computer Science

Gayatri Vidya Parishad College of Engineering

Jul 2014 – May 2018

Bachelor of Technology, Electrical and Electronics Engineering